

Chapter 6

Development Plans Quality Plans

Introduction

OK, we have the **contract**, now to do the **planning**:

We need both a **Development Plan** and a **Quality Plan**

We have proposal plans and internal documents. Enough? No!

These plans typically included time tables, estimates, staffing requirements, scheduled reviews, risks and more.

Seems like enough....

But the **time invested** in developing a development plan and a quality plan will play dividends.

Henry Kissinger once said: if you don't have a plan, any road will get you there....Words to the Wise...

Introduction

- Plans need
 - Proposal materials **re-examined** and **updated** since contract time
 - Much more comprehensive planning with respect to
 - Schedule
 - Resource estimates
 - Development risk evaluations (technical, social, environmental, ...)
 - **Many times there is a considerable lag between the plans and the OK!!**
 - Lots of things take place: staffing, securing budget, manpower, other urgent items...
 - Additional subjects missing from approval proposal (ahead)
 - To include the ability to sound alerts regarding scheduling difficulties, potential staff shortages, scarcity of development facilities, problems meeting contractual milestones, additional risks, etc.
 - We **MUST review** to ensure we are all on the same page.

Introduction

- Development and quality standards (ISO 9000.3 and CMM) require viable plans.
 - We will discuss ISO 9000.3 and CMM and CMMi in considerable detail later.
- We need to look at development plans and quality plans – their objectives and elements.
- They are related but NOT the same.
- Many times quality plans simply do not occur!
 - Often times, if/when they do, they are pie in the sky!

Chapter Objectives

- At the conclusion of this study, you need to be able to:
 - Explain the **objectives** of a development plan and a quality plan
 - Identify the **elements** of a development plan
 - Identify the **elements** of a quality plan
 - Identify the major **software risk items**
 - Explain the process of **software risk management**
 - Discuss the importance of development and quality plans for **small** projects
 - Discuss the importance of development and quality plans for **internal** projects.

Development and Quality Plans - Objectives

Planning is meant to prepare adequate foundations for successful and timely completion of the project. The planning process includes:

1. **Scheduling development activities and estimating the required manpower resources and budget**
2. **Recruiting team members and allocating development resources**
3. **Resolving development risks**
4. **Implementing required SQA activities**
5. **Providing management with data needed for project control**

Elements of the Development Plan

1. **Project Products, specifying “Deliverables”**

Must specify items to be delivered to customer
documents, operations manuals, user manuals,

Must specify specific software products (along with
completion and installation dates
programs (unless centrally controlled), files ...
Address conversion dates; handover, if maintaining;

Must specify or discuss training.
Who and how does training take place?

Must specify customer support!

2. Project Interfaces

Interfaces with existing software packages...

(A course enrollment system might interface with an existing Billing System or Course Scheduling System...)

Interfaces with other software / hardware development and maintenance teams working on similar system or extension of the system

Interface with existing or new hardware

Elements of the Development Plan

3. Project's methodology and development tools

Process used; tools / environment needed

Requirements capture and technologies used

**Design approaches – architectural; procedural;
interface; communications; database....**

Programming methodology

Testing Approaches etc.

**What is the testing responsibilities and who
does what? Individual testing? Separate testing shop?**

Deployment

One shot; parallel; incremental...

4. Software Development Standards and Procedures

Development standards / procedures

(AFDSDCM 3008; AFM 300-4 -Data Elements and Codes)

Elements of the Development Plan

5. Map of the Development Process

Detailed definition of project's phases:
inputs, activities, outputs (artifacts), specific activities

Estimates of each activity's duration:
design reviews (managerial and technical), tests,
design and code correction activities,

Sequencing and dependency of activities

List professional resources needed overall and for each activity

Can show these in GANTT charts, which shows various activities by horizontal bars and whose lengths are proportional to the activity's duration.

Can use PERT and CPM and other activities to communicate activities, durations, deliverable dates, ...

Some like Microsoft Project.

Can use more modern tools too, like IBM's Rational Team Concert (RTC)

6. Project milestones

Often part of the process – e.g. RUP, Scrum, other agile processes or other methodologies.

**Major milestones; Minor milestones (what these mean!)
Often, these are defined by the software process
used.**

What are major and minor milestones in RUP?

**In practice, these are often declared up front and we
work backwards!!**

Elements of the Development Plan

7. Project staff organization and coordination with external participants

Organizational structure – defines teams and tasks;

Defines expertise needed (certifications, experience, specialties), programming languages, development tools, levels of expertise, numbers of individuals needed and for specific periods of time; names of team leaders and team members (sometimes).

Often these responsibilities are ‘shredded out’ too

Long term leadership; team losses due to many factors; Estimates of staff availability is crucial and can cause the flag to be raised when certain levels are not met.

Elements of the Development Plan

8. Required development facilities

Required hardware, software, tools, space, infrastructure, ...

9. Development Risks and Risk Management Actions

Development Risk: “a state or property of a development task or environment, which, if ignored, will increase the likelihood of project failure.

Risk Areas

Technology Risks – lack of expertise; not correct / needed tools

Personnel / Staff Shortages – loss of people; inability to recruit

Environmental Risk:

Where / how is application go be deployed?

Where / how is app to be developed?

Is it an app for the battlefield? Home?

Financial Risks

Social?

Interdependence of other organizational elements who supply resources (subcontractors, specialized

hardware, etc.)

Risk: Risk Identification, Risk Evaluation, Risk Mitigation, Risk Resolution.

Implementation of Risk Management Activities (frequency, threaded....)

Monitoring of Risk Management Activities.

Elements of the Development Plan

10. Control methods

Progress reports and coordination meetings are planned to control project implementation

11. Project Cost Estimates

These are based on proposal cost estimates followed by thorough review and continual updating.

Changes can/will occur and these can be major budget impacts, such as subcontractors don't fulfill their obligations or other unplanned expenditures arise.

Some projects are 'successes' but way over budget

Ultimately, the approval of the development plan will take place within the organization(s).

Elements of a Software Quality Plan

1. List of Quality Goals

Quality refers to how well the software performs its intended tasks, reliability; how often it fails, mean time between failures, etc.

But quantitative measures are usually preferred because more objective assessments can be made during software development and testing.

Quantitative goals can be used to imply qualitative goodness.

Availability of application? (99.5% of time)

Time to recover? (down time?)

Loading / peak loads? (No. of simultaneous transactions?)

Response time?

Quality goals are reflected in major acceptance criteria and are often reflected in the RFP or Software Requirements Specifications (non-functional software requirements)

2. Review Activities

All quality reviews must be planned:
design reviews,
code inspections,
design inspections, etc.

Also: Scope of review activity (coverage?)
Responsibilities of participants
Selection of participants
Type of review:
(technical, managerial, ...)
Procedures - action lists, identify

only, follow up

3. Software Tests

Must contain comprehensive list of planned tests:

unit, integration, system, ... testing

schedule (part of iteration, phase?)

specific procedures (provide software, test plans, test procedures...Do you have test suits?)

Do you have automated test generators??

specific responsibilities (who does what?)

4. Acceptance Tests for software externally developed
Complete set of acceptance tests planned must exist within the qauality plan.
Especially true for externally developed software and should parallel those tests used for interanally developed software too.

Elements of a Software Quality Plan

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5. Configuration Management Plans: tools, procedures and data for version releases

The quality plan must include tools and procedures for configuration management and change management and control.

Configuration and change management will thread the entire development and quality plans

The Quality Plan may be included in the Development Plan or separate document.

Quality Plans are often seen as design review plans, test plans, acceptance testing plans, etc.

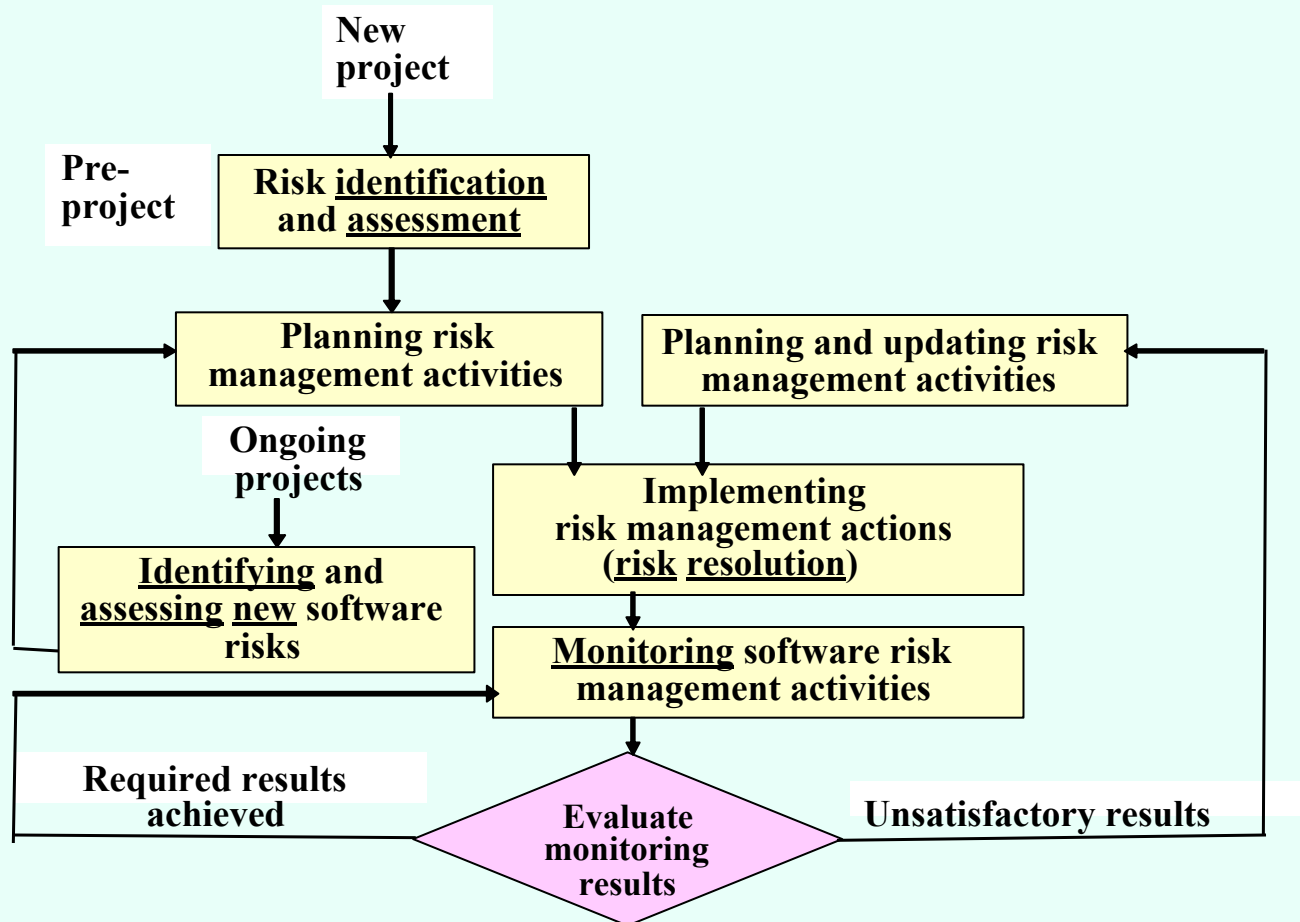
Classes of Software Development Risks

- 1. Scheduling and timing risks**
- 2. System functionality risks**
- 3. Subcontracting risks**
- 4. Requirement management risks**
- 5. Resource usage and performance risks**
- 6. Personnel management risks**

Boehm and Ross's Top 10 Software Risk Items

1. **Developing wrong software functions ***
2. **Unrealistic schedules and budgets *****
3. **Developing wrong user interface**
4. **Gold plating**
5. **Continuing stream of requirement changes *****
6. **Shortfalls in externally furnished components**
7. **Shortfalls in externally performed tasks**
8. **Personnel shortfalls ****
9. **Real-time performance shortfalls**
10. **Straining computer science capabilities**

The Software Risk Management Process



Homework – Chapter 6

- Individually, you are to answer Question 6.4 and turn it in to Blackboard Assignment 6 no later than 4pm next class period.

Team Assignment

- Team 1

See question 6.6 for guidance on what further to discuss.

- You are not required to discuss it all, but....
- Look up /use Appendix 6A and present a comprehensive discussion on many software risks. See pages 112-115.
- This is a great opportunity to do some real work in this key area.